Team Details

a. **Team name:** Lambdac Computing
b. **Team leader name:** Reshma Mohan
c. **Problem Statement:** Problem Statement 4 - Default Prediction Model

Brief about the idea

CreditPulse - Early Warning (Default Prediction Model): default prediction that buys the bank time

- The brief: current default-prediction capability is limited, built on structured repayment data and fragmented methodologies; the outcome sought is stress identified 12 months in advance under a common interpretation framework.
- Our answer: A borrower's alternate-data footprint (GST turnover, UPI inflows, EPFO payroll, e-way bills, electricity consumption) often deteriorates months before repayment behaviour does. CreditPulse monitors that footprint monthly and flags emerging risk even while every EMI is still being paid on time
- The output: a portfolio deterioration radar, a ranked watchlist with explained drivers per borrower, and a calibrated 12-month default probability under a common Red / Amber / Green framework.
- Lead time is evaluated against a repayment-only baseline (the incumbent's own lens): median 11.5 months vs 2.0 months - an 8-month head start.

This entry is one track of CreditPulse, one codebase and one deployment answering three problem statements (PS3 Financial Health Score, PS4 Early Warning / Default Prediction, PS5 Fraud Intelligence). Reviewers land directly on this track at its deep link.

Opportunities

How different is it from any of the other existing ideas?

- Most default-prediction entries re-skin a days-past-due tripwire or ship an unvalidated black-box score. We headline lead time against an explicit repayment-only baseline, not raw accuracy - for early warning, the value is time to act.
- No data leakage, and tested for it: entity-level train/test split, future-window features raise instead of leaking, labels are attached in a separate step.

How will it be able to solve the problem?

- A calibrated 12-month default probability on structured (repayment) plus alternate (GST / UPI / EPFO / e-way / electricity) data, scored monthly across the whole book.
- Every flag is explained - which footprint signals are rolling over - so a relationship manager can act inside the cure window: review limit, site visit, restructure.

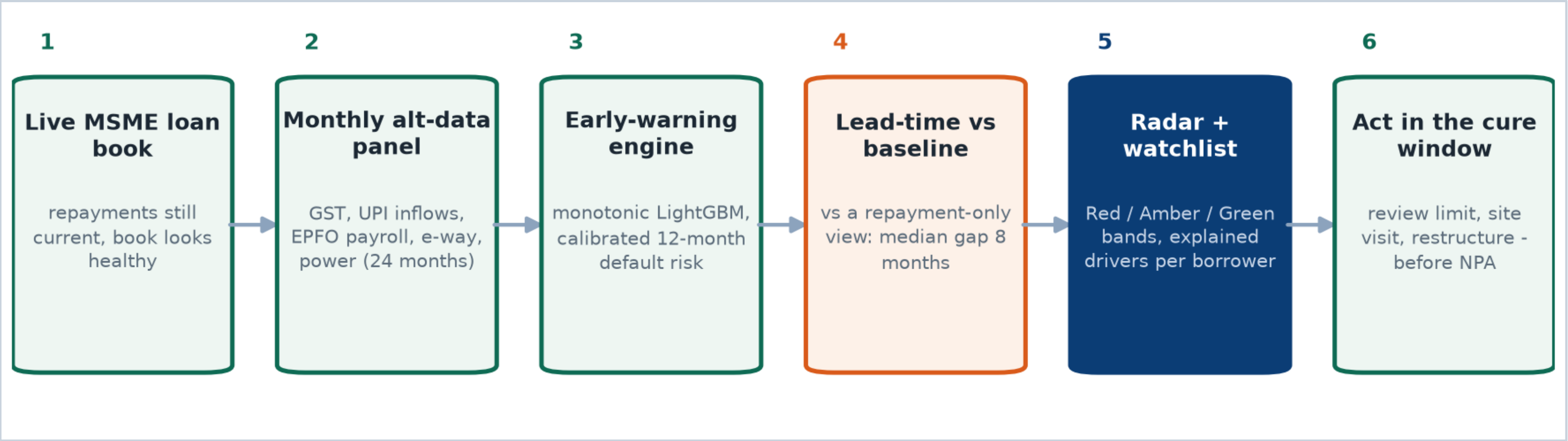
USP of the proposed solution

- A median 8-month head start over a repayment-only view (11.5 vs 2.0 months) and capture@top-decile 0.926 vs 0.519, measured by an in-repo eval harness on a leakage-guarded holdout.

List of features offered by the solution

- Portfolio Overview: a deterioration radar across the live book with Red / Amber / Green bands and exposure-weighted KPIs
- Watchlist: ranked flagged borrowers with 12-month default risk, days late, exposure and a recommended action per row
- Case drilldown: a 24-month timeline of the alt-data footprint vs repayments, with first-alert markers for our model, the baseline and the projected default
- Explained drivers per borrower (GST trending down, payroll shrinking, inflows shrinking) - plain language, defensible to an officer
- Monotonic LightGBM with isotonic calibration; the repayment-only baseline is trained the same way, so the comparison is fair
- Leakage guards enforced in code and tests: entity-level split, future-window features raise, labels attached separately
- Lead time, capture@decile and alert precision as first-class evaluation metrics
- Common interpretation framework shared across the platform: same bands, same explanation style, one deployment
- Batch monthly scoring; stateless engine, pre-fitted and baked into the image

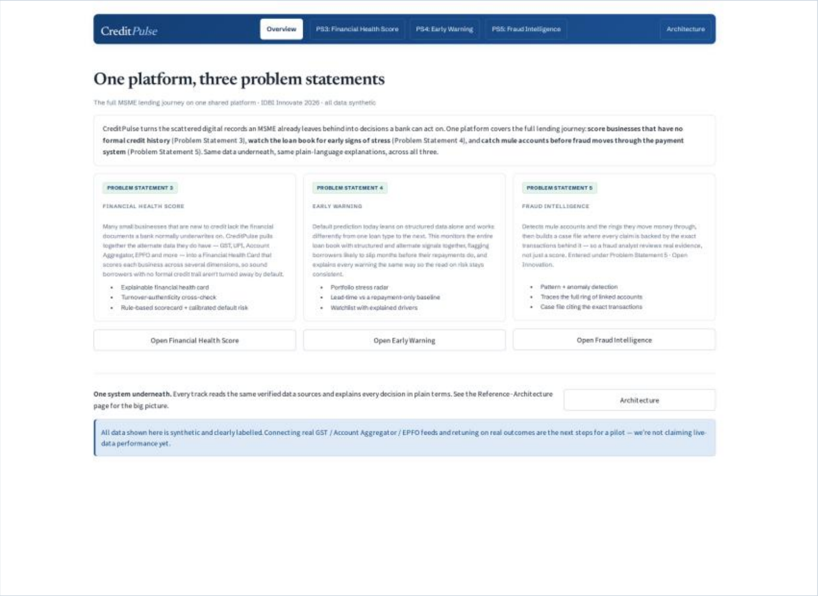
Process flow diagram or Use-case diagram



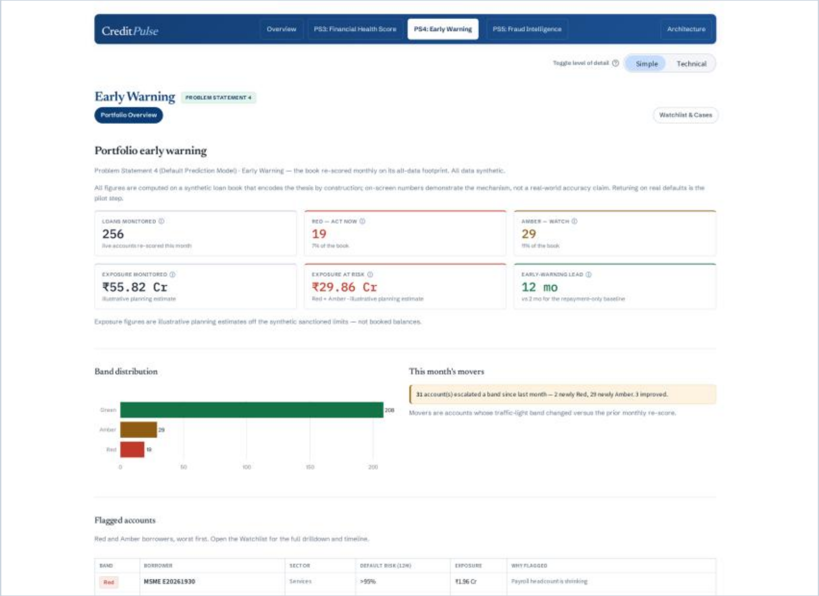
The monthly monitoring loop: the footprint is re-read every month, the engine re-scores the book, and flagged borrowers arrive with their reasons while the cure window is still open.

Wireframes/Mock diagrams of the proposed solution (optional)

The prototype is implemented and deployed; actual product screens below stand in for wireframes.

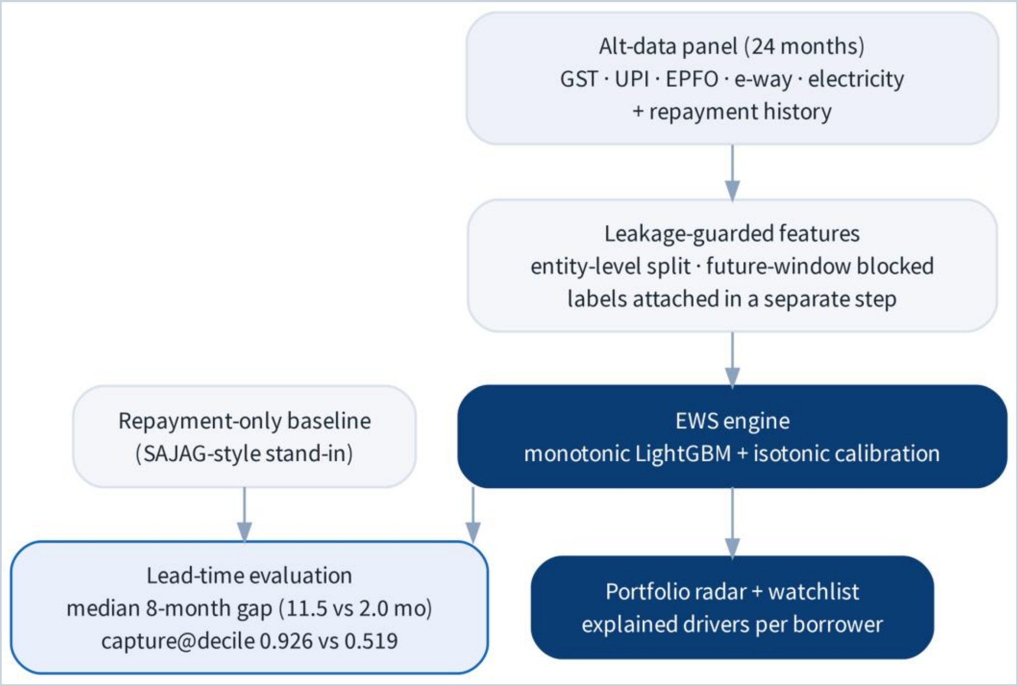


Platform landing page: one card per problem statement



Portfolio Overview: the deterioration radar over the live synthetic book

Architecture diagram of the proposed solution



Rendered live inside the product (Architecture page, Problem Statement 4 view).

Self-contained track

- Everything lives in one folder (data generation, features, engine, pages); it imports the shared platform core and nothing imports it

Anti-leakage design

- Entity-level split; future-window features raise; labels attached in a separate step - all enforced by tests

Deployment

- Same container and deploy as the other tracks; deep link /track04; engine pre-fitted at build time

Baseline discipline

- The repayment-only baseline ships in the same codebase, so the lead-time gap is reproducible by judges

Technologies to be used in the solution

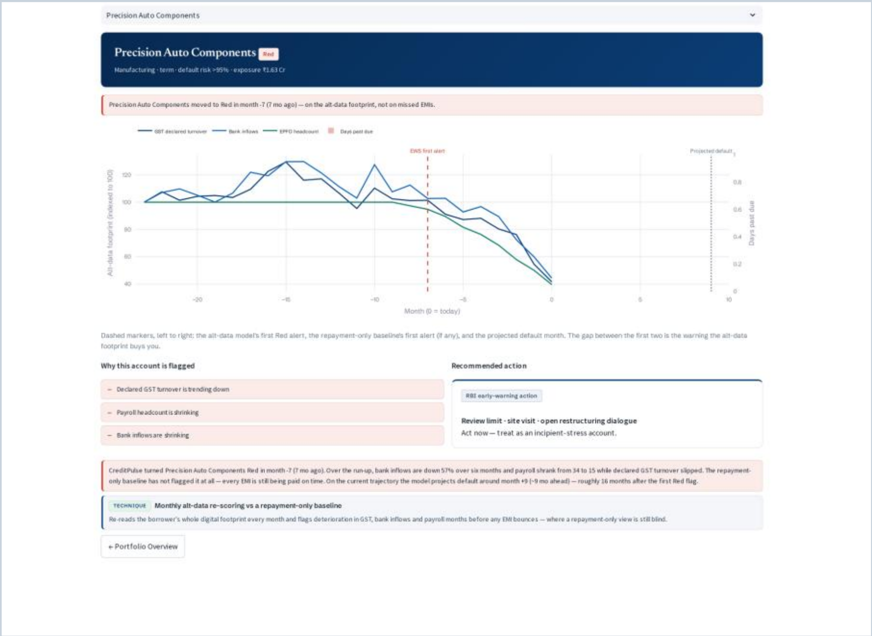
- Python 3.12; pandas + NumPy (data layer), scikit-learn (scorecard, calibration, clustering), LightGBM (monotonicity-constrained gradient boosting), SHAP (explainability)
- Streamlit multi-page product (st.navigation router); Plotly charts; Graphviz architecture diagrams
- Deterministic synthetic-data generators calibrated to public India MSME statistics (RBI, SIDBI-CRISIL, Udyam); fixed seeds, identical output on every run
- Docker single-container deploy (Cloud Run today, portable to the bank's AWS / ACC sandbox at stage 2); GitHub Actions CI regenerates data, runs the full test suite and eval harness, and builds the image on every push
- pytest: 296 tests covering determinism, monotonicity, leakage guards, UI smoke sweeps and track isolation
- No runtime LLM and no black-box service calls: every line of text on screen is assembled from the underlying computation, so the product can be audited end to end
- Track-specific: a 24-month panel generator with cohort construction, repayment simulation and alt-data deterioration arcs driven by shared latent variables

Estimated implementation cost (optional)

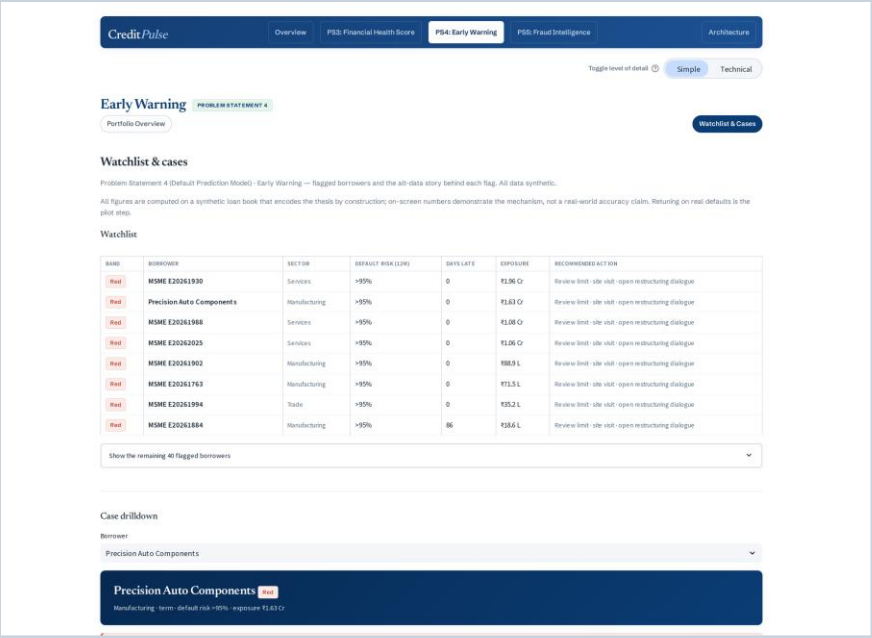
Indicative figures, stage-1 prototype scope.

- Open-source stack end to end: no licence fees, no per-score charges
- One small container (2 vCPU / 2 GB, scale-to-zero) serves all three problem statements; indicative cloud cost well under Rs 5,000 per month at demo traffic
- Pilot-phase cost is engineering, not infrastructure: AA / ULI adapter integration, DPDP consent flows, security review, and model recalibration on bank data
- Scoring is stateless and batch-friendly: portfolio-scale scoring adds compute linearly, with no specialised hardware

Snapshots of the prototype



Case drilldown: the alt-data footprint rolls over at our first alert, months before repayments slip; drivers and recommended action alongside



Watchlist: ranked flagged borrowers with risk, exposure and a recommended action per row

Prototype Performance report/Benchmarking

Eval harness - synthetic holdout, entity-level split

- Median lead time: 11.5 months (ours) vs 2.0 months (repayment-only baseline) - a median 8.0-month gap on paired defaulters
- Capture@top-decile: 0.926 vs 0.519 - the top decile of our score holds 93% of true stress, vs 52% for the baseline
- Holdout AUC 0.973 vs baseline 0.942 (reported for completeness; lead time is the operational value, not accuracy)
- Red-band alert precision 0.855, recall 0.728, false-alert rate 2.9%
- Leakage tests green: future-window features raise; labels attached in a separate step; entity-level split enforced

The synthetic loan book is built to encode this pattern, so on-screen numbers demonstrate the mechanism, not a real-world accuracy claim. Retuning on real defaults is the pilot step.

Additional Details/Future Development (if any)

- Stage 2: backtest on IDBI's historical defaults; recalibrate thresholds per loan type and borrower segment (the common interpretation framework carries over unchanged)
- Add unstructured signals (news, court filings - already catalogued in the platform's data-source catalog) and an account-aggregator refresh cadence
- Integrate with collections and RM workflows: alert routing, action tracking, and cure-rate measurement to close the loop
- Run as a challenger beside the incumbent repayment-based system (SAJAG-style) and measure incremental lead time on live cohorts

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub Public Repository:

https://github.com/lambdacc/idbi

Demo Video Link (3 Minutes):

https://youtu.be/pztpV7OnwQk

Final Product Link:

https://creditpulse-66armtf3tq-el.a.run.app/

Note: On the initial load of the demo deployment, the application may take a few seconds to appear. This is due to an on-demand startup mechanism used to minimize cloud hosting costs during the evaluation period. Once the application has loaded, please select the appropriate Problem Statement tab.

Data honesty and guardrails

- All data in the prototype is synthetic: generated, clearly labelled on screen, and calibrated to public India MSME statistics. Ground-truth labels exist for evaluation only and are never read at score time.
- Built to resist gaming: monotonicity constraints (a worsening risk factor can never improve a score), leakage-resistant holdouts, and determinism tests, all enforced in the 296-test suite
- Every user-visible number is computed from the in-repo eval harnesses at build time, or is labelled illustrative
- Human-in-the-loop: the system recommends, the officer or analyst decides, and every decision leaves an audit trail
- What we do not claim: real-world accuracy. That is measured at pilot, on bank data, after recalibration.

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Build. Integrate. Transform.

Knowledge Partner: aws

Powered by: H2S

Technology Partner: acc applied cloud computing

Thank you!